How to integrate recovery during heat acclimation

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The games of the XXXII Olympiad have been announced from 23 July to 8 August 2021, coinciding with the height of the Japanese summer. Heat acclimation/acclimatisation (HA) is the most important countermeasure athletes can adopt to safeguard their health and performance when competing in the heat. However, it is important for practitioners to recognise that repeated daily exposures to heat stress is a physiologically demanding endeavour. Prolonged heat exposure is characterised by increased thermal and cardiovascular strain, augmented glycogen utilisation and fluid losses, increased hormonal responses (ie, increased cortisol and catecholamine secretion), cytokinaemia and endotoxaemia, which may result in sleep and gastrointestinal disturbances, augmented lethargy and delayed physical recovery.[1–3] For instance, consecutive days of training in the heat, particularly at high exercise intensities, can result in a state of overreaching, despite signs of successful heat adaptation.[4] It is therefore imperative to manage recovery when training in the heat, as accumulated fatigue may impact on the overall success of the HA programme, as well as interfere with the quality of other training sessions. This editorial aims to provide practitioners and coaches with an up-to-date synopsis of recovery strategies during HA (figure 1).

Pre-exercise nutritional strategies are crucial in limiting fluid and electrolyte imbalance during exertional heat stress, such that HA sessions can be optimally engaged while minimising the burden associated with replenishing deficits following exercise. Accordingly, athletes are advised to commence HA sessions euhydrated (ie, within 0.5 kg variation in body mass), with sodium intakes of 0.5–0.7 g/L generally recommended during sessions lasting longer than 60 min.[1 5] The Institute of Medicine has highlighted that public health guidelines regarding sodium intake do not apply to individuals who lose large amounts of sodium through sweat on a daily basis, such as athletes training or competing in the heat.[6] Recovery following heat training may also be influenced by the onset of endotoxaemia and systemic inflammatory response (ie, cytokinaemia), owing to exacerbated splanchic hypoperfusion. In this regard, carbohydrate (CHO) intake prior to (15 g, 6% solution) and during exercise (45 g/hour, 6% solution) has been shown to ameliorate such perturbations through improved splanchnic perfusion, potentially contributing to improved recovery.[7] Overall, athletes are encouraged to commence HA sessions euhydrated and supplement with CHO feedings during exercise, with additional requirement for sodium intake when sessions last over 60 min.

Following exercise, restoration of fluid and glycogen stores are essential, as failure to replenish can impair subsequent training quality. For the complete recovery of fluid balance, it is generally suggested to replace 100%–150% of body mass losses, considering time, tolerance and ongoing urine losses.[1 2] The ingestion of large volumes of fluids in a short period of time (<30 min) should be avoided, with intake spread over 120 min.[2] Sodium is perhaps the most important constituent of a rehydration beverage, and concentrations equal that of the sweat lost seems required to maintain sodium balance and fluid retention.[8] However, there are considerable challenges surrounding sodium recommendation for athletes, given the large interindividual variations in sweat rate (0.16 to >3 L/hour) and sweat sodium content (11.2 to >40 mmol/L or 258 to >920 mg/L), coupled with variations between and within sweat analysis methods.[9 10] Normative values for whole body sweat loss and sweat sodium concentrations have been published to guide practitioners in adapting fluid replacement guidelines and strategies.[9] Nevertheless, the fact remains that the only way to determine an athlete's sodium requirement is to collect and analyse sweat samples when exercising in the heat, following which sodium needs can be estimated by multiplying sweat sodium content by sweat rate.[10]

The inclusion of CHO or protein during recovery will both independently enhance fluid retention. However, beverages (eg, milk) or foods including CHO, protein and sodium are perhaps most suited to meet other recovery goals including electrolyte balance restoration, glycogen replenishment and muscle protein synthesis.[8 11] Athletes undertaking HA do not seem to require additional CHO requirements than that recommended for postexercise recovery in normothermic environments (ie, 1 g/kg/hour or 0.8 g/kg/hour when coingested with protein), although heat stress has been shown to augment muscle glycogenolysis.[12] HA methods (eg, controlled heart rate, controlled hyperthermia and rating of perceived exertion- clamped method) all result in substantially reduced absolute exercise intensities for a given relative intensity, likely resulting in similar overall CHO utilisation between neutral and hot environments. Moreover, CHO oxidation rates in environmental conditions of ~35°C are unlikely to be augmented compared with exercise in cooler conditions at relative intensities below 80% VO_{2max},[13] well below the exercise intensities typically sustained during HA (<70% VO_{2max}).

To minimise the cumulative effects of thermal strain across training sessions, it is advised that athletes rest in cool environments and avoid additional heat exposure, other than that undertaken during HA. On the premise that an adequate HA stimulus of 60–90 min is undertaken,[1] it is imperative that preceding and ensuing hours are focused towards recovery or preparedness for other training sessions. Incorporating head and face cooling may be an option for some athletes following heat training, as such strategies have been shown to minimise the stress response contributing to sensations of fatigue, lethargy and unwillingness to train.[14] Athletes experiencing sleep disturbances during HA should review their sleep hygiene practices and adopt simple strategies such as regular sleep–wake hours, having a comfortable sleep environment (ie, cool, dark and quiet) and avoiding stimulants such as electronic devices (eg, mobile phones and laptops) or caffeine for at least 1 hour and 4–5 hours (might vary between individuals), respectively, before bedtime.[15] Athletes who undertake heat training late in the day and consequently have difficulties falling asleep or experience poor sleep quality can incorporate postexercise cold water immersion (12°C–15°C for

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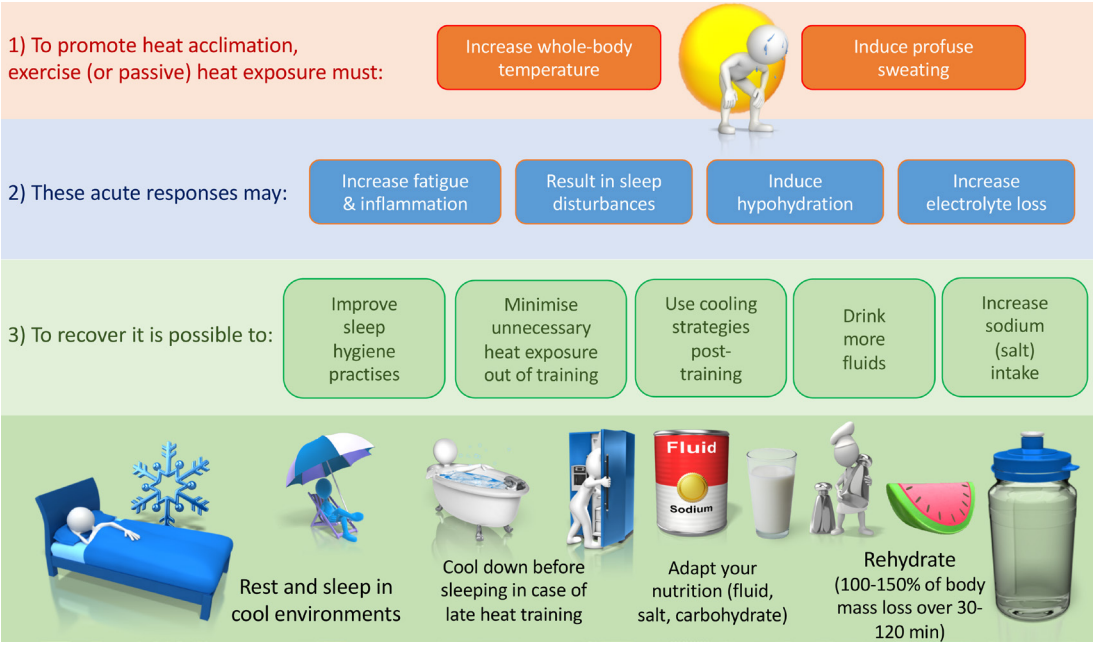

Figure 1 Schematic illustration on the responses to repeated training in the heat and recovery strategies.

10–15 min) as part of their recovery, as this strategy has been shown to minimise the decrease in subjective sleep quality in athletes undertaking heat training.[3]

In summary, it is important to recognise that training in the heat results in substantial physiological perturbations, which need to be managed to minimise accumulated fatigue and improve athletes' readiness to train. Nutritional plans supporting adequate hydration, electrolyte balance and postexercise fuelling are fundamental to the recovery process. Unnecessary heat exposure (ie, other than that undertaken during HA) should be avoided, while recovery and sleep should be undertaken in cool, comfortable environments. If needed, recovery may include head and face cooling or cold water immersion to improve athlete perception and sleep, respectively.

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